Practical - 13

Practical Title: PHP JSON API and AJAX/Fetch-Based CRUD without Page Reload

Problem definition: Create simple PHP JSON API endpoints for events or students and consume them using Fetch API. Perform list, add, update, and delete operations asynchronously without full page reload.

Theory:

1. Do PHP endpoints return valid JSON responses?

   Answer: Yes, PHP endpoints return valid JSON responses using json_encode(). The frontend receives and processes the JSON data.

2. Are GET and POST requests handled properly?

   Answer: Yes, GET requests are used to display student records, while POST requests are used to add, update, and delete student records.

3. Are loading, success, and error states shown on the frontend?

   Answer: Yes, the frontend displays loading messages, success messages after operations, and error messages when something goes wrong.

4. Is input validation performed before database operations?

   Answer: Yes, the program checks that all required fields are filled and that the email address is valid before performing database operations.

A. Implementation/ Output/Result (Keep only one option which is suitable to practical)

Tags Used and Their Purpose

Tag	Purpose
`<html>`	Defines the HTML document.
`<head>`	Contains page information and styles.
`<title>`	Sets the webpage title.
`<style>`	Adds CSS styling to the webpage.
`<body>`	Contains the visible webpage content.
`<h1>`	Displays the main heading.

`<form>`	Creates the student input form.
`<input>`	Accepts student details.
`<button>`	Performs add, update, clear, and delete actions.
`<p>`	Displays loading, success, and error messages.
`<table>`	Displays student records.
`<thead>`	Defines the table headings.
`<tbody>`	Contains student records added dynamically.
`<tr>`	Defines a table row.
`<th>`	Defines a table heading cell.
`<td>`	Defines a table data cell.
`<script>`	Contains JavaScript code for Fetch API and CRUD operations.

B. Evidence of Work (Screenshots)

Student Management — Fetch API

Students loaded successfully.

Student name | Email | BCA | Save Student | Clear

Student List

ID	Name	Email	Course	Actions
3	Rohan Mehta	rohan@example.com	BTech	Edit Delete
2	Diya Patel	diya@example.com	BSc IT	Edit Delete
1	Aarav Shah	aarav@example.com	BCA	Edit Delete

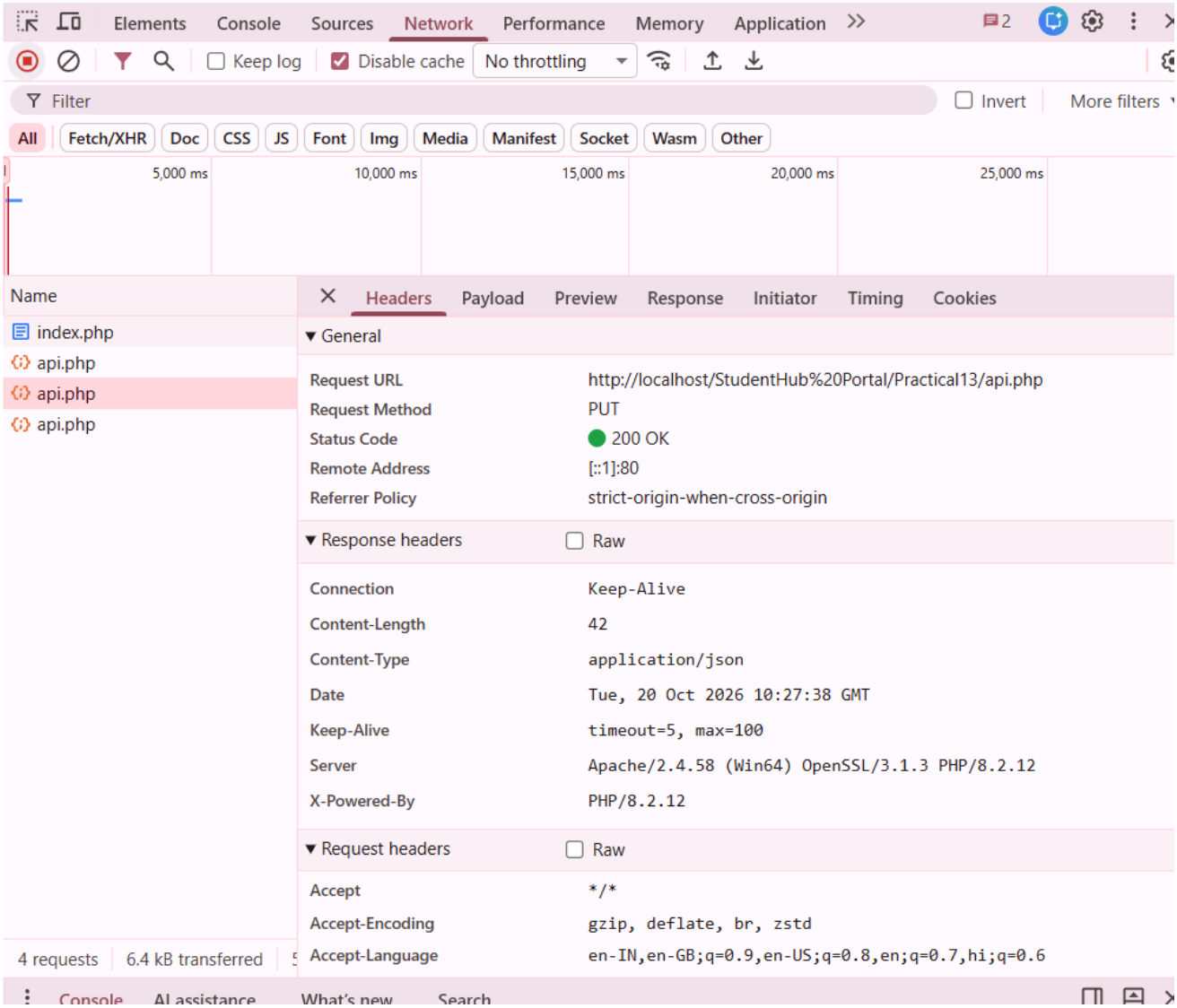
Student Management — Fetch API

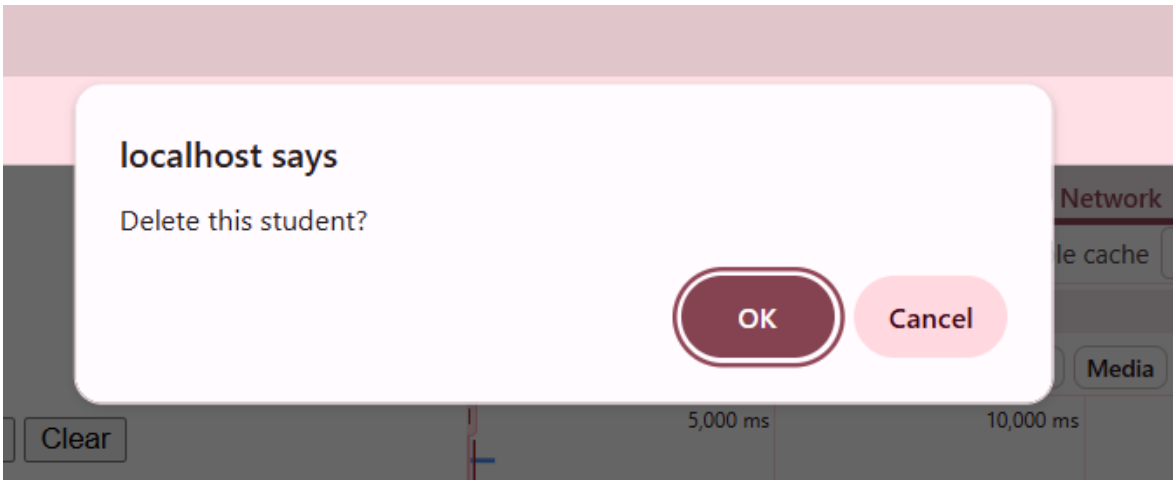
Students loaded successfully.

Student name | Email | BCA | Save Student | Clear

Student List

ID	Name	Email	Course	Actions
4	Heer	heershaah@gmail.com	BTech	Edit Delete
3	Rohan Mehta	rohan@example.com	BTech	Edit Delete
2	Diya Patel	diya@example.com	BSc IT	Edit Delete
1	Aarav Shah	aarav@example.com	BCA	Edit Delete


Elements
Console
Sources
Network
Performance
Memory
Application
Keep log
Disable cache
No throttling
Filter
Invert
More filters
All
Fetch/XHR
Doc
CSS
JS
Font
Img
Media
Manifest
Socket
Wasm
Other
5,000 ms
10,000 ms
15,000 ms
20,000 ms
25,000 ms
Name
index.php
api.php
api.php
api.php
Headers
Payload
Preview
Response
Initiator
Timing
Cookies
General
Request URL
http://localhost/StudentHub%20Portal/Practical13/api.php
Request Method
PUT
Status Code
200 OK
Remote Address
[::1]:80
Referrer Policy
strict-origin-when-cross-origin
Response headers
Raw
Connection
Keep-Alive
Content-Length
42
Content-Type
application/json
Date
Tue, 20 Oct 2026 10:27:38 GMT
Keep-Alive
timeout=5, max=100
Server
Apache/2.4.58 (Win64) OpenSSL/3.1.3 PHP/8.2.12
X-Powered-By
PHP/8.2.12
Request headers
Raw
Accept
/
Accept-Encoding
gzip, deflate, br, zstd
Accept-Language
en-IN,en-GB;q=0.9,en-US;q=0.8,en;q=0.7,hi;q=0.6
4 requests
6.4 kB transferred
Console
AI assistance
What's new
Search


localhost says
Delete this student?
OK
Cancel
Network
Media
Clear
5,000 ms
10,000 ms

C. GitHub Repository Link:

https://github.com/heerce27/StudentHub-Portal.git

D. Challenges Faced:

- Connecting PHP API endpoints with the MySQL database.
- Sending and receiving data using JSON and Fetch API.
- Performing CRUD operations without reloading the webpage.

Learning Outcomes:

- Learned how to create PHP JSON API endpoints.
- Learned how to use Fetch API for asynchronous communication.

Faculty Remarks: